

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

7th September 2018

Subject ST6 – Finance and Investment B

Time allowed: Three Hours (10.15* – 13.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. ** You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You have then three hours to complete the paper.*
3. *You must not start writing your answers in the answer sheet unless instructed to do so by the supervisor.*
4. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
5. *Attempt all questions, beginning your answer to each question on a separate sheet.*
6. *Mark allocations are shown in brackets.*
7. *Please check if you have received complete Question paper and no page is missing. If so, kindly get a new set of Question paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer booklet and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** Richmond Investments Limited has launched a new single premium with profits product, with multiple fund options and initial asset allocation as given below

Initial asset mix	
Equity	50%
C-bond	30%
Government bond	20%

There are also guaranteed payouts of premium at maturity plus any bonus declared during the policy term. The bonus declared is the greater of '75% of the profit in the fund' and 'risk free return for that year'.

The parent company has also invested some cash in the company to back the guarantees.

The balance sheet as on 31/03/2018 for the company is

Balance sheet t=0	
Assets	
Fund value	1200
Company Cash	600
Total assets	1800
Liabilities	
Policyholder Share	1300
Guarantees cost	300
Total Liability	1600
Free capital	200

The risk free rate as used for the next year is 4% per annum, compounded annually.

- i) Explain the two approaches by which the guarantees can be valued by the company. In particular, explain the:
 - Key steps.
 - Mathematical principles.
 - Assumptions regarding inputs, advantages and disadvantages of the model.
 - Key risk drivers and any interaction between them.(7)
- ii) The Regulator has requested the company to have sufficient free capital to withstand 40% fall in the equity market. Construct a balance sheet for 40% fall in equity markets and estimate the amount which is needed to be invested by the company. Assume the cost of guarantee has delta of 12% of fund value.
 (7)
- iii) An investment actuary has suggested using the options market instead of putting more cash from the company. Suggest a hedging strategy such that the balance sheet is immune to a fall in the market and is also cost effective. Consider the following steps in creating the strategy

- Approximate amount of market fall which can be sustained with the existing free capital (within a range of 5%)
- Consider movements of assets and liabilities with hedges in place for both rise and fall of equity allowing for an increase in policyholder benefits (exact figures are not required but you are required to specify key constraints where free capital is impacted)
- State the risks to be considered for these hedges

(7)
[21]

Q. 2) New Government regulations allow individuals to not pay tax on income if the payment is deferred until retirement and then paid to the individual in regular annual payments. The company has designed remuneration for its HR director who is planning to retire in the next 10 years at the rate of Rs. A per annum. The compensation as also determined now should increase in line with RPI subject to a maximum of 5% per annum until the retirement age but the cumulative inflation impact is floored at 0%.

- i) State the formula for the annual payment commencing at the date of retirement and create a replicating portfolio which would have similar annual payment at the date of retirement (do it for just one annual payment).
- ii) If the RPI follows a lognormal distribution with a continuous rate of return of η and interest rate δ , derive a formula for the payment after 10 years.

(6)
(6)
[12]

Q. 3) Zicron Limited decided to buy a 3 year bond at par (to be redeemed at face value on maturity) having a face value of Rs. 1000 with coupon rate of 8%. The bond is given a credit rating of A. The bond default model used by Zicron Limited assumes that the annual default rate is determined by two factors: historical behaviour and the current level of the equity market.

- i) Explain why the current equity market level might impact the default rate of bonds. State the various approaches by which this can be allowed for when estimating the default rate.

An investment actuary has suggested using a style of Capital Asset Pricing Model to calculate the default rate for each year.

The investment actuary has suggested adapting the default rate model which has linear dependency on the equity level.

$$D_a = D_f + \beta(E_m - E_f)$$

where

D_a : Default probability for the bond

D_f : long term default rate

E_m : Current equity volatility level

E_f : long term equity volatility level

β = Model parameter

(4)

- ii) Describe how this model can be adapted to estimate the default probability and suggest an approach to calibrate the model to derive the parameters. State possible limitations of the model.

(5)

- iii) Using the CAPM model, the Investment actuary has calculated the default rate for a 2 scenario equity market: 10% up of 4 % and 10% down of 6%. Using the following inputs, state in which scenario the bond value would be **closer to its par value**.

- recovery rate of 50%
- interest rate of 5% compounded annually.

State any assumptions about the cashflow timing used

(8)

[17]

- Q. 4)** Consider a European call option on a stock with strike price of \$50, time to maturity of 65 days (65/365 years) and volatility of 25% per annum. The current spot price of the stock is \$50 and the risk-free rate of interest in the market is 6% per annum (with continuous compounding).

- i) Calculate the value of the above mentioned European Call option using the Black-Scholes formula. (3)
- ii) Calculate the value of Delta and Gamma for the above mentioned option. (4)
- iii) The writer has sold 1000 calls (the call option sold is the same as mentioned above) and wants to have a delta neutral portfolio at the current time. To ensure delta neutrality,
 - a) How many shares would the writer need to buy? (2)
 - b) How much additional money would the writer need to invest / borrow? (2)
- iv) If the next day the share price moves to \$52 with the other terms remaining constant, what changes would the writer need to make to the hedge portfolio to ensure delta neutrality for the 1000 calls written? (3)
- v) On the day the option is written, the writer wants to be both Delta and Gamma neutral for the 1000 calls written. Incidentally, in trying to attain delta and gamma neutrality, the writer finds that another option (Option 2) on the same stock is available in the market. However, the strike price for Option 2 is \$55 while the time to maturity is (100/365) years. The price for this option is 1.1466; the corresponding delta and gamma for Option 2 are 0.2965 and 0.0529 respectively. In that case,
 - a) How many new options (Option 2) should the writer buy? (2)
 - b) How many shares should the writer buy? (2)
 - c) How much additional money does the writer need to invest / borrow? (1)
 - d) Comment on why the cash requirement is lower in this case as compared to the delta neutral case? (1)

[20]

- Q. 5)**
- i) Connect the Vasicek model with the AR(1) process. (2)
 - ii) For any short rate model $dr = \mu(r, t)dt + \sigma(r, t)dW$ that produces zero coupon bond prices of the form $P(t, T) = A(t, T) e^{-B(t, T)r(t)}$, show that the spot rate volatility structure is the curve $\sigma(r, t)B(t, T)/(T-t)$. (3)

- iii) Suppose that the spot rate curve $r(r, a, b, t, T) = r + a(T - t) + b(T - t)^2$ is implied by a three factor model. Which of the factors, r , a and b affects slope, curvature and parallel moves respectively? (3)
- iv) Between a normal model and log-normal model, which overprices out – of – the money calls on bonds and underprices out – of – the – money puts on bonds and why? (2)
- v) When pricing derivative on bonds under the HJM model, does the tree have to be built over the life of the longer – term underlying bond or just over the life of the derivative? Explain your answer. (3)
- [13]

Q. 6) Let the current risk free spot rate curve be as follows:

Term (years)	Risk free Spot Rate (%)
0.5	0.54
1.0	0.61
1.5	0.73
2.0	0.85
2.5	1.00
3.0	1.15
3.5	1.32
4.0	1.48
4.5	1.64
5.0	1.80
5.5	1.94
6.0	2.08

Note that all interest rates are expressed as continuously compounded per annum. Further, all instruments considered below are payable semi-annually and the respective rates are convertible half-yearly.

- i) What would be the forward swap rate for a 3 year swap starting 3 years from now? (4)

A financial institution is committed to paying 3.5% p.a. interest (convertible half – yearly and payable semi-annually) on a Euro 100 million nominal amount for 3 years from time 3. The investor is likewise committed to depositing Euro 100 million with the financial institution and earning 3.5% p.a. during that period.

The financial institution is considering hedging this deal against interest rate falls. It is considering two alternative hedging deals:

- Entering into a forward rate agreement (FRA) to receive 3.5% p.a. on the Euro 100 million nominal over the three year period from time 3.
- Purchasing an “at – the – money” receiver swaption on a 3 year swap with an exercise date in three years’ time.

The Black volatility for such an “at – the – money” 3 X 3 swaption is 20% p.a.

- ii)** Using this information,
 - a)** Calculate the price of the FRA. (2)
 - b)** State whether the financial institution would pay or receive this amount. (2)
 - iii)** What would be the price of the swaption? (5)
 - iv)** Compare the two hedging approaches. (4)
- [17]**
